Network Printer Configuration

1. Introduction

Network printers allow multiple users on the same network to access and use a single printer device. Instead of connecting a printer to one computer, it is connected to the network so all authorized users can print documents. This project demonstrates how to configure a printer on a network using Cisco Packet Tracer and understand basic concepts like IP addressing and device integration.

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2. Objective

The objectives of this project are:

- To understand how a printer works in a network environment.
- To configure a printer with a proper IP address.
- To connect multiple computers to a network printer.
- To verify successful printing through network connectivity.

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3. Tools Used

The following tool is used in this project:

- Cisco Packet Tracer

Cisco Packet Tracer is a network simulation tool used to design, configure, and test computer networks without requiring real hardware devices.

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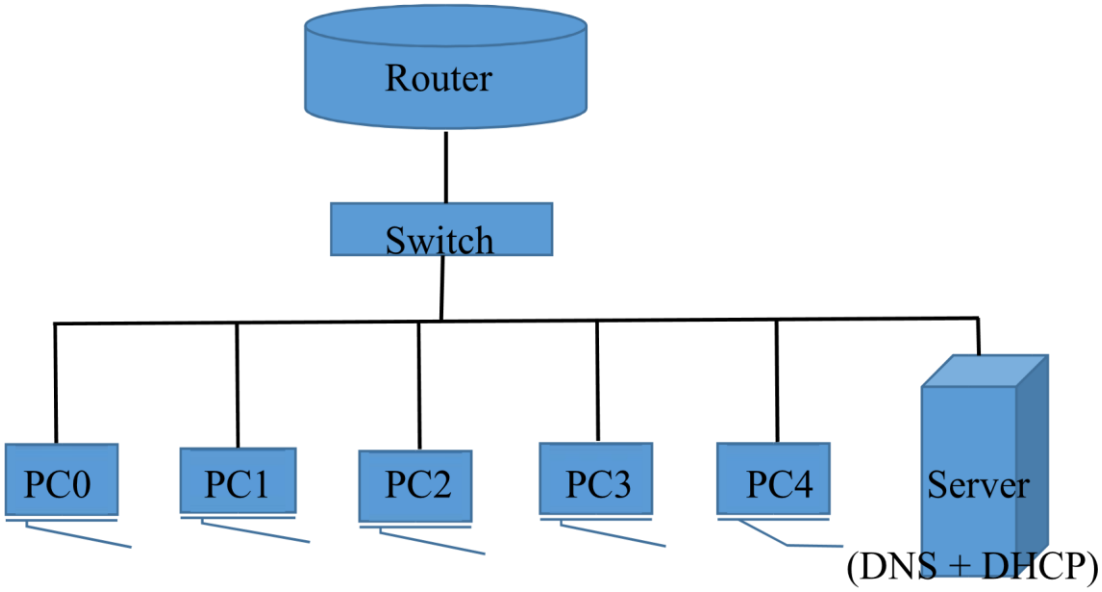
4. Network Devices Used

Device	Model	Purpose

Router	ISR4331	Gateway routing between network segments.
Switch	2960-24TT	Connecting multiple PCs within the office.
Server	---	Dedicated device for DNS and DHCP services.
PC	---	End devices requesting network services.

5. Network Topology

All client computers are connected to a switch using Copper StraightThrough cables. The switch is connected to a router which connects the network to the server.



(Network Structure)

6. IP Address Configuration

Device/Pool	IP Address	Subnet Mask	Purpose
Router	192.168.1.1	255.255.255.0	Default Gateway
Server	192.168.1.3	255.255.255.0	DNS+DHCP Service
Switch	192.168.1.2	255.255.255.0	Switch Management
PC	DHCP	255.255.255.0	Client
anudip.in	192.168.1.10	255.255.255.0	-
d2216.com	192.168.1.20	255.255.255.0	-

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7. Network Implementation Steps

Step 1: Network Design

• Add **Router, Switch, Printer, and PCs** in Cisco Packet Tracer.
• Connect devices using **Copper Straight-Through cables**.

Step 2: Printer Configuration

1. Click on the printer device.
2. Go to **Config > FastEthernet**.
3. Assign the following IP address:

IP Address: **192.168.1.10**
Subnet Mask: **255.255.255.0**
Default Gateway: **192.168.1.1**

This makes the printer accessible on the network.

Step 3: PC Configuration

1. Open each PC.
2. Go to **Desktop > IP Configuration**.
3. Assign IP addresses manually or using DHCP.

Example:

PC0 → 192.168.1.20
PC1 → 192.168.1.21

Step 4: Printer Access Setup

1. Open the **PC Desktop**.
2. Go to **Printer Settings / Network Printing**.
3. Enter the printer IP address **192.168.1.10**.
4. Send a test print request.

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8. Network Testing

After configuration, the network is tested to ensure that the printer works properly.

Connectivity Testing Use

command prompt in PC:

ping 192.168.1.10

If the printer responds, the network connection is successful.

Printing Test
Send a **test document** from a PC to the printer.
If the document prints successfully, the configuration is correct.

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9. Conclusion

This project demonstrates how to configure a **network printer** in a LAN environment using Cisco Packet Tracer. By assigning an IP address and connecting the printer to the network switch, multiple computers can share the same printer efficiently. Network printing reduces hardware costs and improves resource sharing in organizations.